Biochemical and Biophysical Research Communications 683 (2023) 149090

Contents lists available at ScienceDirect

Biochemical and Biophysical Research Communications

journal homepage: www.elsevier.com/locate/ybbrc

Unveiling the peptidome diversity of *Lachesis muta* snake venom: Discovery of novel fragments of metalloproteinase, L-amino acid oxidase, and bradykinin potentiating peptides

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ARTICLE INFO

Keywords:
Lachesis muta
Venom
Peptidomics
Mass spectrometry
Bradykinin potentiating peptides
Snake venom metalloproteinases

ABSTRACT

Snake venoms are known to be major sources of peptides with different pharmacological properties. In this study, we comprehensively explored the venom peptidomes of three specimens of *Lachesis muta*, the largest venomous snake in South America, using mass spectrometry techniques. The analysis revealed 19 main chromatographic peaks common to all specimens. A total of 151 peptides were identified, including 69 from a metalloproteinase, 58 from the BPP-CNP precursor, and 24 from a L-amino acid oxidase. To our knowledge, 126 of these peptides were reported for the first time in this work, including a new SVMP-derived peptide fragment, Lm-10a. Our findings highlight the dynamic nature of toxin maturation in snake venoms, driven by proteolytic processing, post-translational modifications, and cryptide formation.

1. Introduction

Lachesis muta is the largest venomous snake in South America, inhabiting the rainforests from Nicaragua to Brazil [1,2]. In Brazil, the snake is popularly known as "surucucu" [2], which in the Tupi-Guarani language means "the one that bites many times". Accidents involving snakes of the genus *Lachesis* are rare but can be life-threatening [1]. *Lachesis muta* venom has lower toxicity and lethality compared to other venomous species, however, the severity of accidents is attributed to the large amount of venom inoculated [3]. Symptoms observed in patients bitten by *Lachesis muta* include nausea, vomiting, abdominal colic, diarrhea, sweating, hypotension, bradycardia and shock [4].

The venom composition of *Lachesis muta* has been investigated using biochemical methods [5,6], transcriptomics [7], and by the snake venomics approach [8]. These studies have revealed that the venom primarily consists of Zn-metalloproteinases (SVMP), serine proteinases (SVSP), C-type lectins, PLA_2 and bradykinin potentiating peptide/C-type natriuretic peptides (BPP/CNP) [7,8]. SVMPs and SVSPs are responsible for the hemorrhagic and coagulopathic effects of the *L. muta* venom [7], while BPPs potentiate the hypotensive effect of circulating bradykinin [9]. Five BPPs were identified in two cDNA isoforms of *L. muta* venom gland and had their presences confirmed in the venom by tandem mass spectrometry sequencing [9]. Another peptide present in the cDNA precursor was found to be an antagonist of bradykinin action, and it was named bradykinin inhibitory peptide (BIP) [10]. Additionally, several other peptides have been discovered in *L. muta* and other *Lachesis* sp. venoms [8,11,12].

Venom peptidomes exhibit complex compositions and are rich sources of bioactive molecules [13,14]. Previous studies have demonstrated that venom peptidomes are not only rich in BPPs, but also contain substantial amounts of native peptides derived from other venom toxins such as SVMPs, LAAOs, disintegrins and other fragments of the BPP/CNP precursor [14,15]. While the proteome and transcriptome of *L. muta* venom have been comprehensively explored [7,8], the peptidome remains underinvestigated to date. In this study, we collected venoms from three specimens of *Lachesis muta* using protease inhibitors and characterized their venom peptidomes through a combination of liquid chromatography coupled with mass spectrometry, database search and *de novo* analysis.

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https://doi.org/10.1016/j.bbrc.2023.10.022
Received 29 September 2023; Accepted 5 October 2023
Available online 6 October 2023
0006-291X/

2. Materials and methods

2.1. Venoms

Venoms were collected from three specimens of *Lachesis muta*: two (one male and one female) from São José da Lage (AL, Brazil) and one (male) from Rio Branco (AC, Brazil). The specimens were kept in the Laboratório de Herpetologia, Instituto Butantan (SP, Brazil), and venoms were collected as previously described [13,14]. Briefly, venoms were collected into a beaker in ice bath, immediately mixed with an inhibitor solution (PMSF 2 mM and EDTA 5 mM), subjected to a centrifugation at 1700 g for 30 min at 4 °C to remove debris, lyophilized and stored at −20 °C. The procedures were performed in accordance with the relevant guidelines and regulations. Venoms were analyzed under the authorization for access to the genetic patrimony, SisGen number A57EBEE. All procedures were approved by the Research Ethics Committees of Universidade Federal de São Paulo and Instituto Butantan (Protocol number: CEUA n^{o} 2589220322 and CEUAIB n^{o} 7967310720).

2.2. Mass spectrometry (MS)

Lyophilized samples were dissolved in formic acid 0.1% and submitted to a solid-phase extraction (SPE) using C18 stage tips [16]. Peptides were eluted with 40% ACN, as previously described [15]. The eluted peptidomics samples were vacuum-dried and stored at −20 °C until MS analysis. MS analysis was performed on a Synapt G2 HDMS mass spectrometer (Waters) coupled to a nanoAcquity UPLC system (Waters). *L. muta* peptidomics samples were dissolved in 0.1% formic acid (FA). The aliquots were injected into a trap column (Acquity UPLC M-Class Symmetry C18 Trap Column, 100 Å, 5 μm, 300 μm × 25 mm, Waters) for the desalinization with phase A (0.1% formic acid) at a flow rate of 8 μl/min for 5 min, and then eluted to an analytical column (Acquity UPLC M-Class HSS T3 Column, 1.8 μm, 300 μm × 150 mm, Waters) using an elution gradient of 7–35% B (0.1% formic acid in ACN) in 60 min at 3 μl/min. The mass spectrometer was operated in data-independent acquisition (DIA) and data-dependent acquisition (DDA) modes as previously described [17,18]. The ESI source was operated in positive mode with capillary voltage of 3.0 kV, lock temperature of 100 °C, and cone voltage of 40 V. Venoms of each specimen were analyzed in technical duplicates, resulting in a total of 12 LC-MS/MS runs, 6 for DIA and 6 for DDA. The mass spectrometry data have been deposited to the ProteomeXchange Consortium via the PRIDE [19] partner repository with the dataset identifier PXD045732.

2.3. Bioinformatics

Data were processed and analyzed as previously described [17]. Briefly, for DIA runs, MS data was processed in Progenesis QI for proteomics (PQIP, Nonlinear Dynamics) using low energy threshold of 750 counts and elevated energy threshold of 50 counts. Data were normalized and aligned with a reference run automatically selected. After processing, a mgf file with all MS/MS spectra was exported from PQIP and imported into PEAKS Studio 7.5 (Bioinformatics Solutions Inc.) for *de novo* analysis, database search, post-translational modification (PTM) and mutation (SPIDER) searches. The *de novo* sequenced peptides with average local confidence (ALC) scores ≥ 50% were selected for database searches. The processing parameters in PEAKS Studio 7.5 were as follows: mass tolerance of 10 ppm for precursor ions, mass tolerance of 0.025 Da for fragment ions, no enzyme specificity, N-terminal acetylation, methionine oxidation and pyroglutamic acid from N-terminal glutamic acid or glutamine as variable modifications, and maximum of three PTMs per peptide. Database, PTM and SPIDER searches were performed using UniProt database of Lachesis (180 sequences, downloaded on November 12th, 2021), utilizing *de novo* sequenced peptides and a maximum false discovery rate of 1%. For the DDA runs, MS raw data were processed in the PEAKS Studio X+ (Bioinformatics Solutions Inc.) using the same parameters as PEAKS Studio 7.5.

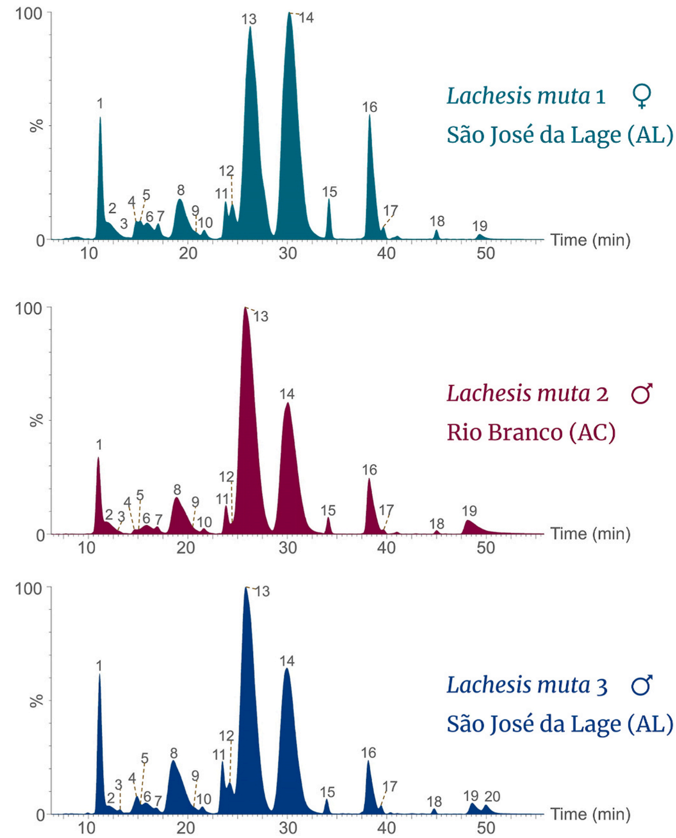

Fig. 1. RP-HPLC chromatogram of peptides from the venom of three specimens of *Lachesis muta* obtained by solid phase extraction, highlighting the higher intensity peaks.

2.4. Manual de novo analysis

Peptides not identified by automated searches were subjected to manual *de novo* analysis, as previously described [13,14]. Peptides were validated if the observed masses were within 30 ppm of the theoretical mass, and major fragments observed in MS/MS spectra matched b or y ions, with a minimum of five matches [18].

3. Results and discussion

3.1. Analysis of L. muta venom peptidomes

Analysis of the LC-MS/MS profiles indicates that the three *Lachesis muta* venom peptidomes are similar. Nineteen main chromatographic peaks were present in all three venoms (Fig. 1 and Table 1) and a total of 151 peptides were identified in the three specimens, comprising 69 from SVMPs, 58 from BPP-CNP, 24 from LAAOs (Supp. Table 1). Among these, 25 peptides had previously been reported in other studies, while, to our knowledge, 126 new peptides are reported for the first time in this work (Supp. Table 1). It is worth noting that peaks 17 and 18 (Fig. 1) presented peptide ions with insufficient MS/MS fragments for confident identification.

The peptides BIP [10], Lm-BPP 1, Lm-BPP 2, Lm-BPP 3, Lm-BPP 4 and Lm-BPP 5 [9] were identified in peaks 1, 14, 8, 13, 2 and 16, respectively (Fig. 1 and Table 1). Curiously, these peptides originate from the same BPP-CNP precursor [9] and are among the most intense ions of the peptidome, collectively representing 66.9% of total ion intensities (Table 1). Additional 19 peptides, mainly BPP-CNPs, have been

Table 1
Lachesis muta venom peptides identified in the main chromatographic peaks of LC-MS/MS analysis.

Mass (Da)	RT[a] (min)	Peptide Sequence	Protein Accession	Protein family	LC-MS peak	$\log_{10}(Lm_1)$[b]	$\log_{10}(Lm_2)$[b]	$\log_{10}(Lm_3)$[b]	Av. I[c] (%)	Description	References
1078.54	10.92	TP(hyp) PAGPDVGPR	Q27J49	BPP-CNP	1	5.35	5.29	5.25	0.46%	Hyp-BIP[d]	This work
1062.54	10.92	TPPAGPDVGPR	Q27J49	BPP-CNP	1	6.11	6.09	6.30	3.41%	BIP	[8,10,11]
1191.63	11.79	KYNGNLNTIR	P22796	SVMP	2	5.96	5.89	5.84	1.83%	–	This work
1403.74	11.93	ZKKWPPGHHIPP	Q27J49	BPP-CNP	2	6.17	6.09	6.03	2.91%	Lm-BPP 4	[8,9,11, 12]
558.25	13.20	ZKWD	Q27J49	BPP-CNP	3	4.66	4.83	4.98	0.15%	*de novo*, Lm-BPP 5[g]	This work
752.38	14.62	TPPAGPDV	Q27J49	BPP-CNP	4	5.26	5.00	5.05	0.29%	BIP[g]	This work
928.45	15.12	ZKPWPPGH	Q27J49	BPP-CNP	5	5.57	5.33	5.69	0.79%	Lm-BPP 2[g]	[12]
1036.51	15.80	WPPGHHIPP	Q27J49	BPP-CNP	6	5.94	5.88	5.93	1.93%	Lm-BPP 2[g]	[11,12]
465.27	16.40	PPRP	Q27J49	BPP-CNP	6	5.31	5.07	5.29	0.39%	*de novo*, Lm-BPP 1[g]	This work
651.34	16.47	WPPRP	Q27J49	BPP-CNP	6	5.43	5.20	5.39	0.51%	Lm-BPP 1[g]	This work
906.44	16.93	TPPAGPDVGP	Q27J49	BPP-CNP	7	5.50	5.46	5.25	0.59%	BIP[g]	This work
1372.69	18.99	ZKPWPPGHHIPP	Q27J49	BPP-CNP	8	6.54	6.64	6.80	10.62%	Lm-BPP 2	[9]
1035.54	19.22	LKTFGEWR	P22796	SVMP	8	5.03	5.01	5.10	0.26%	–	This work
832.34	20.58	ZEWPPGH	Q27J49	BPP-CNP	9	5.71	5.63	5.75	1.16%	Lm-BPP 3[g]	This work
849.40	21.62	ZKWDPPP	Q27J49	BPP-CNP	10	5.38	5.31	5.40	0.54%	Lm-BPP 5[g]	This work
703.39	23.78	PPPISPP	Q27J49	BPP-CNP	11	5.67	5.71	5.91	1.34%	Lm-BPP 5[g]	This work
734.38	24.62	ZKPWPP	Q27J49	BPP-CNP	12	5.95	5.68	6.04	1.80%	Lm-BPP 2[g]	This work
1106.53	24.91	LKHDDIFGY	J7H670	LAAO	13	4.91	4.85	4.80	0.17%	–	This work
818.41	25.22	DPPPISPP	Q27J49	BPP-CNP	13	5.83	5.78	5.84	1.52%	Lm-BPP 5[g]	This work
1276.59	25.63	ZEWPPGHHIPP	Q27J49	BPP-CNP	13	6.95	7.10	7.06	25.11%	Lm-BPP 3	[8,9,11, 12]
1405.78	25.76	VGIVQDHSPKTLL	P22796	SVMP	13	4.99	5.09	5.25	0.30%	–	This work
803.46	30.06	PRPQIPP	Q27J49	BPP-CNP	14	5.54	5.54	5.52	0.80%	Lm-BPP 1[g]	This work
1086.59	30.06	WPPRPQIPP	Q27J49	BPP-CNP	14	6.86	6.84	6.85	16.40%	Lm-BPP 1	[8,9,11, 12]
1049.52	31.58	ZKWDPPPIS	Q27J49	BPP-CNP	14	5.48	5.32	5.43	0.60%	Lm-BPP 5[g]	[8]
1293.68	34.13	ZRFSQKYIEL	P22796	SVMP	15	5.60	5.54	5.46	0.79%	*de novo*, Lm-10a	This work
1243.64	38.11	ZKWDPPPISPP	Q27J49	BPP-CNP	16	6.63	6.53	6.52	8.45%	Lm-BPP 5	[8,9] [[11]
1998.90	39.70	–	–	–	17	5.36	5.11	5.44	0.47%	Not identified	This work
1982.91	44.97	–	–	–	18	5.33	5.12	5.24	0.40%	Not identified	This work
13992.4[e]	48.74	HLLQF[f]	–	PLA2	19	6.78	6.64	6.45	9.75%	*de novo*/ homology	[8]
13906.8[e]	50.12	HLLQF[f]	–	PLA2	20	–	–	5.25	0.42%	*de novo*/ homology	[8]

[a] Retention time.
[b] log10 of average peptide ion intensities of each L. muta specimen.
[c] Average ion intensity (of the three replicates) as a percentage of the total ion intensities.
[d] hyp - hydroxyproline.
[e] Average mass.
[f] First five amino acids.
[g] non-canonical processing of a reported peptide.

reported in snake venomics studies [8,11,12](Supp. Table 1). The main peptides identified in the 19 chromatographic peaks account for 94.1% of total ion intensities (Table 1). Furthermore, a PLA_2 with average mass of 13,992.4 eluted in peak 19 (Supp. Fig. S1). The first 5 N-terminal amino acids, HLLQF (Supp. Fig. S1), were obtained through manual *de novo* sequencing of top-down MS/MS spectrum and homology inference with PLA_2 toxins reported by Sanz et al. [8]. Peak 20 also corresponded to a PLA_2 (13,906.8 Da), however, this proteoform was only present in the venom of *L. muta* 03 (Fig. 1, Supp. Fig. S2). PLA_2 is in fact a small protein, but eluted in the peptide fraction of SPE (40% ACN), similarly to BATXPLA8 in the analysis of *Bothrops atrox* venom peptidome [15].

3.2. Identification of a new SVMP peptide fragment

Interestingly, we identified a new SVMP peptide fragment in peak 15: ZRFSQKYIEL (Supp. Fig. S3), named Lm-10a. The sequence was obtained through manual *de novo* sequencing and validated using database search. The peptide contains a pyroglutamic acid (Z) at the N-terminus and the amino acids Q, K, I and L were inferred by homology to the SVMP LHFII (UniProt: P22796), whose sequence was obtained through Edman degradation in the pioneering work of Sanchez et al. [20]. The first two amino acids, ZR, were missing from LHFII in the original sequence, but we hypothesize that the SVMP undergoes partial proteolytic processing, and both forms coexist in the venom gland. The identification of the peptide FSQKYIEL in the venom peptidome (Supp. Table 1) supports our hypothesis. Additionally, other homologous sequences from other venomous snakes have QR at corresponding positions [20]. We have recently described a new SVMP-derived cryptide (a peptide with distinct function from its putative parent protein [21]) from the venom of *Bothrops cotiara*, Bc-7a, whose sequence is ZRYDPFR [22]. Bc-7a is a novel angiotensin converting enzyme (ACE) inhibitor, while the SVMP-III from which it is derived is a hemorrhagic protease [23,24]. Notably, Bc-7a and Lm-10a share the first two amino acids (ZR) and derive from the N-terminal region of a SVMP. Lm-10a may also be a new ACE inhibitor, potentially characterizing it as a new cryptide. However, further experimental work will be required to confirm this potential activity.

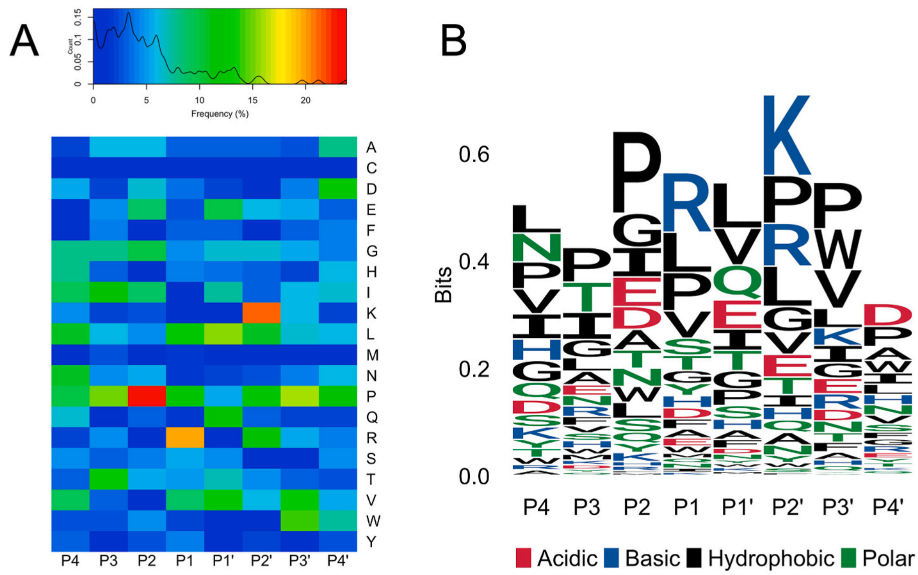

Fig. 2. Amino acid cleavage frequencies at positions P4–P4′ of peptides identified in the venom peptidome of *Lachesis muta*. A) Amino acid frequencies represented in the heatmap; B) Amino acid frequencies represented in the sequence logo graphs.

3.3. Complexity of venom proteomes and peptidomes

Proteolytic analysis of *L. muta*, based on amino acid frequencies at positions P4–P4' (Supp. Table 1 and Fig. 2), reveals a high prevalence of Pro at positions P2 (24%) and P1 (13%), consistent with the compositions of BPPs [14,25]. The high abundance of Pro, along with the diversity of proteolytic processing sites, results in the widespread presence of this amino acid across the P4–P4′ range (Fig. 2). Gln at P1' (12%) is also relatively frequent, likely attributable to the high content of BPP-CNP peptides in *L. muta* venom. Arg emerges as the most frequent residue at position P1' (20%), in line with prior studies that have analyzed toxin processing in *Bothrops* sp. snakes [14,22,26]. This finding further underscores the significance of this residue in Viperidae toxin proteolytic processing. Additionally, we have identified peptides that map a Leu-His mutation in LHFII (L48H, Supp. Table 1).

The venoms were collected with protease inhibitors and the main peptides have been consistently identified in the three venoms, strongly indicating that the peptides undergo maturation in the venom glands and may play important roles in envenomation. Non-canonical BPP-CNP, SVMP, LAAO and disintegrin fragments have also been found in other snake venom peptidomics and proteomics studies [8,11,12,14,15,27]. New cryptides, proteolytic processing, post-translational modifications and mutations contribute to the complexity of venom proteomes and peptidomes. This dynamic nature of toxin maturation demonstrates that venomous snakes have employed various biological mechanisms over evolution to refine their venoms. Despite the advancement of omics technologies and the generation of large amounts of sequencing data in recent years, a vast universe of peptides and their biological roles remain to be discovered in snake venoms.

CRediT authorship contribution statement

Lucas T. Ito: Formal analysis, Investigation, Methodology, Writing – original draft, Writing – review & editing. **Jackson G. Miyamoto:** Data curation, Formal analysis, Validation, Writing – review & editing. **Sávio S. Sant'Anna:** Methodology, Resources, Writing – review & editing. **Kathleen F. Grego:** Methodology, Resources, Writing – review & editing. **Anita M. Tanaka-Azevedo:** Conceptualization, Methodology, Resources, Writing – review & editing. **Alexandre K. Tashima:** Conceptualization, Formal analysis, Funding acquisition, Methodology, Project administration, Resources, Software, Supervision, Validation, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This research was funded by Fundação de Amparo à Pesquisa do Estado de São Paulo (grants 2017/20106-9, 2019/07629-8, 2017/21052-0 and 2021/11707-4), Financiadora de Estudos e Projetos (FINEP), Conselho Nacional de Desenvolvimento Científico e Tecnológico (CNPq, 140689/2019-2 to JGM) and Coordenação de Aperfeiçoamento de Pessoal de Nível Superior – Brasil (CAPES) - Finance Code 001.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.bbrc.2023.10.022.

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